

# Dr. Megan Masterson

NHFP Einstein Fellow at Harvard University

[mmasters@mit.edu](mailto:mmasters@mit.edu) • [memasterson.github.io](https://github.com/memasterson) • ORCID: [0000-0003-4127-0739](https://orcid.org/0000-0003-4127-0739)

## Employment

|                                                                     |           |
|---------------------------------------------------------------------|-----------|
| NASA Hubble Fellowship Program Einstein Fellow, Harvard University  | 2026–2029 |
| Harvard College Observatory Postdoctoral Fellow, Harvard University | 2029–2031 |

## Education

|                                                                                                                                                                                                                                            |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>PhD in Physics, Massachusetts Institute of Technology</b><br><i>Thesis: A Multi-Wavelength Exploration of Transient Accretion onto Supermassive Black Holes</i><br><i>Supervisor: Professor Erin Kara</i>                               | 2020–2026 |
| <b>MASt in Astrophysics, Churchill College, University of Cambridge</b><br><i>Thesis: Extended Fe K<math>\alpha</math> Emission in Nearby AGN Revealed by Multi-Order Chandra HETG Data</i><br><i>Supervisor: Professor Chris Reynolds</i> | 2019–2020 |
| <b>BS in Astronomy, BS in Mathematics &amp; Physics, Case Western Reserve University</b><br><i>Graduated summa cum laude</i>                                                                                                               | 2015–2019 |

## Awards & Scholarships

|                                                                                                                                                                                    |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Graduate Student Poster Award, 20th HEAD Meeting                                                                                                                                   | 2023      |
| Kavli Graduate Fellowship, MIT Kavli Institute for Astrophysics and Space Research                                                                                                 | 2020–2021 |
| Gates Cambridge Scholarship, MASt in Astrophysics, University of Cambridge                                                                                                         | 2019–2020 |
| Chambliss Astronomy Achievement Student Award, 233rd AAS Meeting                                                                                                                   | 2019      |
| Jason J. Nassau Prize, Department of Astronomy, Case Western Reserve University<br><i>To an outstanding senior student in astronomy</i>                                            | 2019      |
| Patricia B. Kilpatrick Award, Case Western Reserve University<br><i>To the four-year varsity athlete with the highest GPA</i>                                                      | 2019      |
| Richard F. Sigal Award, Department of Physics, Case Western Reserve University<br><i>For demonstrating excellence in their studies and intending to pursue a career in physics</i> | 2018      |

## Publications & Astronomer's Telegrams

ORCID: [0000-0003-4127-0739](https://orcid.org/0000-0003-4127-0739), ADS Library: <https://tinyurl.com/mastersonADS>

9 first (or co-first) author publications, 34 co-author publications, 2 ATels  
 $h$ -index = 16, total citations = 850+

### First Author Publications

- [9] **Masterson, M.**, Kara, E., et al. 2025, *Persistence of the Millihertz X-ray Quasi-Periodic Oscillation in the Active Galactic Nucleus 1ES 1927+654*, submitted to ApJ. [arXiv:2604.16688](https://arxiv.org/abs/2604.16688)
- [8] Chakraborty\*, J., **Masterson\***, M., Mummery, A., et al. 2025, *X-ray Spectral-Timing Properties of Tidal Disruption Events*, accepted for publication in ApJ. [arXiv:2602.16868](https://arxiv.org/abs/2602.16868) (\*indicates co-first authors)
- [7] **Masterson, M.**, De, K., et al. 2025, *JWST's First View of Tidal Disruption Events: Compact, Accretion-Driven Emission Lines & Strong Silicate Emission in an Infrared-selected Sample*, ApJL, 988, 2, L48. [doi:10.3847/2041-8213/ade153](https://doi.org/10.3847/2041-8213/ade153)
- [6] **Masterson, M.**, Kara, E., et al. 2025, *Millihertz Oscillations Near the Innermost Orbit of a Supermassive Black Hole*, Nature, 638, 370–375. [doi:10.1038/s41586-024-08385-x](https://doi.org/10.1038/s41586-024-08385-x)
- [5] **Masterson, M.**, De, K., et al. 2024, *A New Population of Mid-Infrared-Selected Tidal Disruption Events: Implications for Tidal Disruption Event Rates and Host Galaxy Properties*, ApJ, 961, 211. [doi:10.3847/1538-4357/ad18bb](https://doi.org/10.3847/1538-4357/ad18bb)
- [4] **Masterson, M.**, Kara, E., et al. 2023, *Unusual Hard X-ray Flares Caught in NICER Monitoring of the Binary SMBH Candidate AT2019cuk/Tick Tock/SDSS J1430*, ApJL, 945, L34. [doi:10.3847/2041-8213/acbea9](https://doi.org/10.3847/2041-8213/acbea9)

- [3] **Masterson, M.**, McDonald, M., et al. 2023, *Evidence for AGN-Regulated Cooling in Clusters at  $z \sim 1.4$ : A Multi-Wavelength View of SPT-CL J0607-4448*, ApJ, 944, 164. [doi:10.3847/1538-4357/acae9e](https://doi.org/10.3847/1538-4357/acae9e)
- [2] **Masterson, M.** & Reynolds, C.S. 2022, *Probing the Extent of Fe K $\alpha$  Emission in Nearby AGN Using Multi-Order Analysis of Chandra High Energy Transmission Grating Data*, ApJ, 936, 66. [doi:10.3847/1538-4357/ac83ae](https://doi.org/10.3847/1538-4357/ac83ae)
- [1] **Masterson, M.**, Kara, E., et al. 2022, *Evolution of a Relativistic Outflow and the X-ray Corona in the Extreme Changing-Look AGN 1ES 1927+654*, ApJ, 934, 35. [doi:10.3847/1538-4357/ac76c0](https://doi.org/10.3847/1538-4357/ac76c0)

#### Co-Author Publications.....

- [34] Sánchez-Sáez, P., **Masterson, M.**, et al. 2026, *Spatially resolved optical and mid-infrared spectroscopy of SDSS1335+0728: implications for the origin of the Ansky event*, submitted to A&A. [arXiv:2607.00921](https://arxiv.org/abs/2607.00921)
- [33] Sadaula, D. et al. (including **Masterson, M.**) 2026, *The emergence of X-ray emission lines during relativistic radio-jet formation in the changing-look active galactic nucleus 1ES 1927+654*, accepted for publication in ApJ. [arXiv:2607.05246](https://arxiv.org/abs/2607.05246)
- [32] Nair, P., Panagiotou, C., **Masterson, M.**, et al. 2026, *A Suppressed Volumetric Rate of High-luminosity Mid-infrared Selected Tidal Disruption Events*, ApJ, 1007, 1, 40. [doi:10.3847/1538-4357/ae83a1](https://doi.org/10.3847/1538-4357/ae83a1)
- [31] Vega, J. et al. (including **Masterson, M.**) 2026, *Ghosts of eruptions past: Searching for historical Galactic supernovae using variable thermal dust echoes and machine learning*, submitted to ApJ. [arXiv:2604.08686](https://arxiv.org/abs/2604.08686)
- [30] Wang, M. et al. (including **Masterson, M.**) 2026, *Testing the Association of Supermassive Black Hole Infrared Flares and High-energy Neutrinos*, ApJL, 998, 1, L29. [doi:10.3847/2041-8213/ae3f90](https://doi.org/10.3847/2041-8213/ae3f90)
- [29] De, K. et al. (including **Masterson, M.**) 2026, *Fading into darkness: A weak mass ejection and low-efficiency fallback accompanying black hole formation in M31-2014-DS1*, submitted to ApJL. [arXiv:2601.05774](https://arxiv.org/abs/2601.05774)
- [28] Berger, V. et al. (including **Masterson, M.**) 2026, *Disk-to-Corona State Transition and Extreme X-ray Variability in the Tidal Disruption Event AT2019teq*, accepted in ApJ. [arXiv:2601.04311](https://arxiv.org/abs/2601.04311)
- [27] Zaidouni, F. et al. (including **Masterson, M.**) 2026, *XRISM finds the Changing-Look AGN NGC 1365 in an extended low state: A dense, highly ionized outflow obscures the central source*, ApJL, 997, 1, L26. [doi:10.3847/2041-8213/ae307d](https://doi.org/10.3847/2041-8213/ae307d)
- [26] Neights, E. et al. (including **Masterson, M.**) 2026, *GRB 250702B: Discovery of a Gamma-Ray Burst from a Black Hole Falling into a Star*, MNRAS, 545, 2, staf2019. [doi:10.1093/mnras/staf2019](https://doi.org/10.1093/mnras/staf2019)
- [25] Golay, W. et al. (including **Masterson, M.**) 2026, *Radio Emission from the Infrared Tidal Disruption Event WTP14adeqka: The First Directly Resolved Delayed Outflow from a TDE*, ApJ, 1007, 1, 88. [doi:10.3847/1538-4357/ae8536](https://doi.org/10.3847/1538-4357/ae8536)
- [24] Baldini, P. et al. (including **Masterson, M.**) 2025, *A new Bowen Fluorescence Flare and Extreme Coronal Line Emitter discovered by SRG/eROSITA*, A&A, 701, A224. [doi:10.1051/0004-6361/202554812](https://doi.org/10.1051/0004-6361/202554812)
- [23] Arcodia, R. et al. (including **Masterson, M.**) 2025, *SRG/eROSITA No. 5: Discovery of quasi-periodic eruptions every  $\sim 3.7$  days from a galaxy at  $z > 0.1$* , ApJ, 989, 1, 13. [doi:10.3847/1538-4357/ade9b](https://doi.org/10.3847/1538-4357/ade9b)
- [22] Grotova, I. et al. (including **Masterson, M.**) 2025, *The population of tidal disruption events discovered with eROSITA*, A&A, 697, A159. [doi:10.1051/0004-6361/202553669](https://doi.org/10.1051/0004-6361/202553669)
- [21] Chakraborty, J. et al. (including **Masterson, M.**) 2025, *Rapidly varying ionization features in a Quasi-periodic Eruption: a homologous expansion model for the spectroscopic evolution*, ApJ, 984, 2, 124. [doi:10.3847/1538-4357/adb972](https://doi.org/10.3847/1538-4357/adb972)
- [20] Chakraborty, J. et al. (including **Masterson, M.**) 2025, *Discovery of Quasi-periodic Eruptions in the Tidal Disruption Event and Extreme Coronal Line Emitter AT2022upj: implications for the QPE/TDE fraction and a connection to ECLs*, ApJL, 983, 2, L39. [doi:10.3847/2041-8213/adc2f8](https://doi.org/10.3847/2041-8213/adc2f8)
- [19] Yao, Y. et al. (including **Masterson, M.**) 2025, *Distinguishing Tidal Disruption Events and Changing-look Active Galactic Nuclei via Variation of Mid-infrared Color*, ApJS, 281, 1, 7. [doi:10.3847/1538-4365/ae047a](https://doi.org/10.3847/1538-4365/ae047a)
- [18] Earl, N. et al. (including **Masterson, M.**) 2025, *AT 2020nov: Evidence for Disk Reprocessing in a Rare Tidal Disruption Event*, ApJ, 983, 1, 28. [doi:10.3847/1538-4357/adb974](https://doi.org/10.3847/1538-4357/adb974)
- [17] Rani, B. et al. (including **Masterson, M.**) 2025, *High-Frequency Power Spectrum of AGN NGC 4051 Revealed by NICER*, ApJL, 981, 1, L18. [doi:10.3847/2041-8213/adace8](https://doi.org/10.3847/2041-8213/adace8)

- [16] Grotova, I. et al. (including **Masterson, M.**) 2025, *eRO-ExTra: eROSITA extragalactic non-AGN X-ray transients and variables in eRASS1 and eRASS2*, A&A, 693, A62. [doi:10.1051/0004-6361/202451253](https://doi.org/10.1051/0004-6361/202451253)
- [15] Laha, S. et al. (including **Masterson, M.**) 2025, *Multi-wavelength observations of a jet launch in real time from the post-changing-look Active Galaxy 1ES 1927+654*, ApJ, 981, 2, 125. [doi:10.3847/1538-4357/adaea0](https://doi.org/10.3847/1538-4357/adaea0)
- [14] Meyer, E. et al. (including **Masterson, M.**) 2025, *Emergence of a radio jet in the changing-look AGN 1ES 1927+654*, ApJL, 979, 1, L2. [doi:10.3847/2041-8213/ad8651](https://doi.org/10.3847/2041-8213/ad8651)
- [13] García, J. et al. (including **Masterson, M.**) 2024, *The high energy X-ray probe (HEX-P): science overview*, FrASS, 11, 1471585. [doi:10.3389/fspas.2024.1471585](https://doi.org/10.3389/fspas.2024.1471585)
- [12] De, K. et al. (including **Masterson, M.**) 2024, *The disappearance of a massive star marking the birth of a black hole in M31*, accepted for publication in Science, [arXiv:2410.14778](https://arxiv.org/abs/2410.14778)
- [11] Li, R. et al. (including **Masterson, M.**) 2024, *The Interplay between the Disk and Corona of the Changing-look Active Galactic Nucleus 1ES 1927+654*, ApJ, 975, 140. [doi:10.3847/1538-4357/ad7aed](https://doi.org/10.3847/1538-4357/ad7aed)
- [10] Zhang, Z. et al. (including **Masterson, M.**) 2024, *Modeling X-Ray Multi-Reflection in Super-Eddington Winds*, ApJ, 977, 2, 157. [doi:10.3847/1538-4357/ad86c0](https://doi.org/10.3847/1538-4357/ad86c0)
- [9] Myers, C. et al. (including **Masterson, M.**) 2024, *WTP19aalnxx: Discovery of a bright mid-infrared transient in the emerging class of low luminosity supernovae revealed by delayed circumstellar interaction*, ApJ, 976, 230. [doi:10.3847/1538-4357/ad8922](https://doi.org/10.3847/1538-4357/ad8922)
- [8] Pasham, D. et al. (including **Masterson, M.**) 2024, *A Case for a Binary Black Hole System Revealed via Quasi-Periodic Outflows*, Science Advances, 10, 13. [doi:10.1126/sciadv.adj8898](https://doi.org/10.1126/sciadv.adj8898)
- [7] Wang, Y. et al. (including **Masterson, M.**) 2024, *Rapid dimming followed by a state transition: a study of the highly variable nuclear transient AT 2019avd over 1000+ days*, ApJ, 962, 78. [doi:10.3847/1538-4357/ad182b](https://doi.org/10.3847/1538-4357/ad182b)
- [6] Kammoun, E., Lohfink, A. M., **Masterson, M.**, et al. 2024, *The High Energy X-ray Probe (HEX-P): Probing the physics of the X-ray corona in active galactic nuclei*, FrASS, 10, 1308056. [doi:10.3389/fspas.2023.1308056](https://doi.org/10.3389/fspas.2023.1308056)
- [5] Brightman, M. et al. (including **Masterson, M.**) 2024, *The High Energy X-ray Probe (HEX-P): Sensitive broadband X-ray observations of transient phenomena in the 2030s*, FrASS, 10, 1292656. [doi:10.3389/fspas.2023.1292656](https://doi.org/10.3389/fspas.2023.1292656)
- [4] Panagiotou, C., De, K., **Masterson, M.**, et al. 2023, *A Luminous Dust-Obscured Tidal Disruption Event Candidate in a Star Forming Galaxy at 42 Mpc*, ApJL, 948, L5. [doi:10.3847/2041-8213/acc02f](https://doi.org/10.3847/2041-8213/acc02f)
- [3] Kara, E. et al. (including **Masterson, M.**) 2023, *UV/Optical disk reverberation lags despite a faint X-ray corona in the AGN Mrk 335*, ApJ, 947, 62. [doi:10.3847/1538-4357/acbcd3](https://doi.org/10.3847/1538-4357/acbcd3)
- [2] Xu, Y. et al. (including **Masterson, M.**) 2022, *Ejection-Accretion Connection in NLS1 AGN 1H 1934-063*, MNRAS, 513, 1910. [doi:10.1093/mnras/stac1058](https://doi.org/10.1093/mnras/stac1058)
- [1] Chakraborty, J., Kara, E., **Masterson, M.**, et al. 2021, *Possible X-ray Quasi-Periodic Eruptions in a Tidal Disruption Event Candidate*, ApJL, 921, L40, ApJL, 921, L40, [doi:10.3847/2041-8213/ac313b](https://doi.org/10.3847/2041-8213/ac313b)

## ATels.....

- [2] Hamada, R., et al. (including **M. Masterson**) 2024, *PRIME discovery of a heavily reddened classical nova PRIME24aadwvh at the Galactic center*, [ATel#16824](https://arxiv.org/abs/2408.16824)
- [1] Pasham, D., et al. (including **M. Masterson**) 2022, *AT2019cuk/SDSSJ1430/ZTF18aarippg: High-cadence NICER and NuSTAR X-ray observations of the potential supermassive black hole binary with imminent merger (the tick-tock source)*, [ATel#15225](https://arxiv.org/abs/2208.15225)

## Accepted Observing Proposals (as PI in bold)

Total grant money obtained as PI: \$341.4k

|                                                                                                              |      |
|--------------------------------------------------------------------------------------------------------------|------|
| <b>Swift (AO22)</b> , 52 ks, \$40k, The QPO-Corona Connection in the Enigmatic AGN 1ES 1927+654              | 2025 |
| <b>Swift (AO22)</b> , 116 targets, 232 ks, fill-in proposal, A Late-Time X-ray View of Mid-IR TDE Candidates | 2025 |
| <b>XMM-Newton (AO25)</b> , 82 ks, \$85.5k, Probing the Origin of the mHz X-ray QPO in 1ES 1927+654           | 2025 |
| <b>JWST (AO4)</b> , co-PI, 3.6 hours, \$80k, Mid-IR overview of turn-on AGN and X-ray QPE SDSS1335+0728      | 2025 |
| <b>NICER (AO7)</b> , 104 ks, \$38.5k, X-ray QPOs and the corona-jet connection in 1ES 1927+654               | 2024 |
| <b>Swift (AO21)</b> , 116 targets, 232 ks, fill-in proposal, A Late-Time X-ray View of Mid-IR TDE Candidates | 2024 |

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| <b>XMM-Newton (AO24)</b> , 30 ks, Constraining the Central Engine & Dust Geometry in Mid-IR TDEs         | 2024      |
| <b>IRTF (2024B)</b> , 16.5 hours, SpeX Spectroscopy of eROSITA-detected Mid-IR Nuclear Flares            | 2024      |
| <b>IRTF (2024A)</b> , 22.5 hours, SpeX spectroscopy of luminous IR echoes of extreme accretion phenomena | 2024      |
| <b>XMM-Newton (AO23)</b> , 5 × 30 ks, Late-Time X-ray Emission in Mid-IR TDEs                            | 2023      |
| <b>JWST (AO2)</b> , 19 hours, A Population of Hidden TDEs with JWST                                      | 2023      |
| <b>NICER (AO5)</b> , 52 ks (NICER), 26 ks (Swift), \$43k, Post-Outburst State of 1ES 1927+654            | 2022      |
| <b>XMM-Newton (AO21)</b> , 70 ks (XMM-Newton & NuSTAR), \$54.4k, Post-Outburst State of 1ES 1927+654     | 2021      |
| <b>XMM-Newton DDT Observations</b> , Total of 315 ks over 11 observations                                | 2024–     |
| <b>Swift ToO Observations</b> , Total of 77 ks over 18 observations                                      | 2021–2026 |
| <b>NICER ToO Observations</b> , Total of 19 ks over 4 observations                                       | 2021–2025 |
| <b>Magellan (MIT Internal Review)</b> , Total of 12 nights over 5 semesters                              | 2023–2025 |

## Ground-Based Observing

|                                                   |           |
|---------------------------------------------------|-----------|
| Magellan Clay/LDSS3, 8 nights, In person & remote | 2023-2024 |
| IRTF/SpeX, 12 nights, Remote                      | 2023-2025 |
| Magellan Baade/FIRE, 4 nights, In person & remote | 2023-2025 |
| Magellan Baade/IMACS, 3 nights, Remote            | 2024      |
| Magellan Baade/MagE, 1 night, Remote              | 2023      |
| 1m class telescopes, > 10 nights, In person       | 2022-2024 |

## Talks & Presentations

*Summary: 28 talks (20 invited, 8 contributed)*

|                                                                                                    |      |
|----------------------------------------------------------------------------------------------------|------|
| <b>Invited Talks</b> .....                                                                         |      |
| High Energy Astrophysical Transients (HEAT) Seminar, University of Maryland, College Park, MD      | 2026 |
| CIERA Observer Group Meeting, Northwestern University, Evanston, IL                                | 2026 |
| Carnegie Lunch Seminar, Carnegie Observatories, Pasadena, CA                                       | 2025 |
| KIPAC Tea Talk, Stanford Kavli Institute for Particle Astrophysics and Cosmology (KIPAC), Virtual  | 2025 |
| MPE High-Energy Seminar, Max Planck Institute for Extraterrestrial Physics (MPE), Virtual          | 2025 |
| Theoretical Astrophysics Center (TAC) Seminar, University of California at Berkeley, Berkeley, CA  | 2025 |
| Friday Lunch Seminar, University of Colorado at Boulder, Boulder, CO                               | 2025 |
| Institute for Fundamental Sciences Seminar, University of Oregon, Eugene, OR                       | 2025 |
| X-ray Quasi-Periodic Eruptions and Repeating Nuclear Transients, ESAC, Madrid, Spain               | 2025 |
| AXIS TDAMM Science Working Group Community Day, Virtual                                            | 2025 |
| <a href="#">UMBRELA</a> Dialogues, Center for Astrophysics, Cambridge, MA                          | 2025 |
| IASF Seminar, INAF/IASF-Palermo, Virtual                                                           | 2024 |
| Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era, Crete, Greece          | 2024 |
| Towards a Physical Understanding of Tidal Disruption Events, UCSB KITP Workshop, Santa Barbara, CA | 2024 |
| 21st HEAD Meeting TDE Special Session, Horseshoe Bay, TX                                           | 2024 |
| Boston Area Black hole Accretion Meeting (BABAM), Center for Astrophysics, Cambridge, MA           | 2023 |
| AAS HEAD Frontiers Seminar, Virtual                                                                | 2023 |
| Astro Seminar, Tufts University, Department of Physics & Astronomy, Somerville, MA                 | 2023 |
| CfA Seminar, Center for Astrophysics, Cambridge, MA                                                | 2023 |
| Extreme Astrophysics Seminar, University of Michigan, Department of Astronomy, Virtual             | 2022 |



|                                                                                                  |      |
|--------------------------------------------------------------------------------------------------|------|
| <b>Contributed Talks</b> .....                                                                   |      |
| Caltech Seminar, Caltech, Pasadena, CA                                                           | 2025 |
| Transients from Space, Space Telescope Science Institute, Baltimore, MD                          | 2025 |
| 245th AAS Meeting, National Harbor, MD                                                           | 2025 |
| Anticipating the Rising Tide of Tidal Disruption Events, UCSB KITP Conference, Santa Barbara, CA | 2024 |
| Black holes Across Space and Time (BLAST) Workshop, Virtual                                      | 2022 |
| COSPAR 2022, 44th Scientific Assembly, Virtual                                                   | 2022 |
| XMM-Newton Workshop: Black Hole Accretion Under the X-ray Microscope, ESAC, Madrid, Spain        | 2022 |
| 19th HEAD Meeting, Pittsburgh, PA                                                                | 2022 |

|                                                      |      |
|------------------------------------------------------|------|
| <b>Contributed Posters</b> .....                     |      |
| 20th HEAD Meeting, Waikaloa, HI                      | 2023 |
| <i>Awarded Graduate Student Poster Award</i>         |      |
| 233rd AAS Meeting, Seattle, WA                       | 2019 |
| <i>Awarded Chambliss Undergraduate Student Prize</i> |      |

## Research Supervision

|                                                                                                            |           |
|------------------------------------------------------------------------------------------------------------|-----------|
| <b>Bernard Leal, MIT MSRP Undergraduate Student (co-supervised with Joheen Charkraborty)</b>               | 2025      |
| Multi-Wavelength Analysis of the Changing-Look AGN RHS 10                                                  |           |
| <b>Taisiia Karasova, MIT Undergraduate Research Student</b>                                                | 2024–2025 |
| Host Galaxy Properties of a New Population of Infrared-Selected Tidal Disruption Events                    |           |
| <b>Kylee Carden, MIT Undergraduate Research Student (co-supervised with Erin Kara)</b>                     | 2022–2023 |
| Probing the Circumnuclear Environment and Ionized Wind in the Changing-Look AGN NGC 1365 with Chandra HETG |           |
| <i>Currently an Astrophysics PhD student at Ohio State</i>                                                 |           |
| <b>Isabella Guilherme, MIT MSRP Undergraduate Student (co-supervised with Erin Kara)</b>                   | 2021–2022 |
| Investigating the Late-Time X-ray Emission in the Tidal Disruption Event AT2019azh                         |           |
| <i>Currently a Physics PhD student at Caltech</i>                                                          |           |

## Teaching

|                                                                                                                               |             |
|-------------------------------------------------------------------------------------------------------------------------------|-------------|
| Subject Design Teaching Certificate, MIT's Teaching + Learning Lab                                                            | Spring 2026 |
| <i>Program that taught how to design a college-level course and culminated in the creation of a learner-centered syllabus</i> |             |
| Mentor, 8.02 (Introductory Electricity and Magnetism)                                                                         | Spring 2025 |
| <i>Mentored 2 students, assisting with problem sets, exam preparation, and time management skills</i>                         |             |
| TA for 12.411 (Astronomy Field Camp), senior undergraduate course at MIT                                                      | Jan. 2024   |
| <i>Assisted with observations for three weeks at Teide Observatory in Tenerife, Spain</i>                                     |             |
| TA for 12.411 (Astronomy Field Camp), senior undergraduate course at MIT                                                      | Jan. 2023   |
| <i>Assisted with observations for three weeks at Teide Observatory in Tenerife, Spain</i>                                     |             |
| Mentor, MIT Physics Directed Reading Program                                                                                  | Jan. 2023   |
| <i>Supervised student in 3 week project learning the basics of black hole accretion physics</i>                               |             |
| TA for 8.287 (Observational Techniques in Optical Astronomy), senior undergraduate course at MIT                              | Fall 2022   |
| <i>Supervised weekly observing, assisted with data reduction and analysis, gave a specialty lecture on X-ray astronomy</i>    |             |
| <b>Student Evaluation Score: 6.8/7</b>                                                                                        |             |
| Teacher for Educational Studies Program, MIT                                                                                  | 2021        |
| <i>Designed and taught two classes on black holes, designed for middle school and high school students</i>                    |             |

## Professional Service

|                                                                                                 |      |
|-------------------------------------------------------------------------------------------------|------|
| <b>Departmental Service</b> .....                                                               |      |
| Chair, Graduate Student Search Committee, MIT Kavli Institute Faculty Hiring Committee          | 2025 |
| Graduate Student Representative, MIT Kavli Institute Public Engagement Officer Hiring Committee | 2024 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Collaborations &amp; Working Groups</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |
| Community Member, LISA Consortium                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025–      |
| Member, LSST Transients and Variable Stars Collaboration, TDE Working Group                                                                                                                                                                                                                                                                                                                                                                                                            | 2025–      |
| Member, Roman Science Collaboration, Explosive Transients Science Group                                                                                                                                                                                                                                                                                                                                                                                                                | 2025–      |
| Member, HEX-P Coronal Physics Working Group                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2023–2024  |
| Graduate Student Member, NICER Observatory Science Working Group                                                                                                                                                                                                                                                                                                                                                                                                                       | 2021–2025  |
| <b>Peer Review</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |
| Referee, <i>Astrophysical Journal</i> , <i>Astrophysical Journal Letters</i>                                                                                                                                                                                                                                                                                                                                                                                                           | 2023–      |
| <b>Conference Organization</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |
| SOC Member, 25th Anniversary of XMM-Newton, Baltimore, MD                                                                                                                                                                                                                                                                                                                                                                                                                              | 2025       |
| <b>Selected Press Coverage</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |
| Star-Shredding Black Holes Hiding in Dusty Galaxies, Masterson et al., <i>ApJL</i><br>Covered by <a href="#">MIT News</a> , <a href="#">Columbia News</a> , <a href="#">Science News</a> , <a href="#">CBS Eyes on the World Podcast</a>                                                                                                                                                                                                                                               | 2025       |
| Millihertz Oscillations Near the Innermost Orbit of a Supermassive Black Hole, Masterson et al., <i>Nature</i><br>Featured in a press conference at the 245th AAS Meeting in January 2025 – watch <a href="#">here!</a><br>Covered by <a href="#">NASA</a> , <a href="#">ESA</a> , <a href="#">MIT News</a> , <a href="#">Scientific American</a> , <a href="#">USA Today</a> , <a href="#">Reuters</a> , <a href="#">Smithsonian Magazine</a> , <a href="#">Swedish National TV</a> + | 2025       |
| A New Population of Mid-Infrared-Selected Tidal Disruption Events, Masterson et al., <i>ApJ</i><br>Covered by <a href="#">MIT News</a> , <a href="#">the Boston Globe</a> , <a href="#">Newsweek</a> , <a href="#">AAS Nova</a> , +                                                                                                                                                                                                                                                    | 2024       |
| <b>Advocacy &amp; Outreach</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |
| <b>Advocacy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |
| Grads Advising Grad Admissions Committee Member, MIT Physics Graduate Student Council<br>Co-lead of organization during 2024-2025 academic year                                                                                                                                                                                                                                                                                                                                        | 2022–      |
| Mentor, MIT Physics Graduate Application Assistance Program                                                                                                                                                                                                                                                                                                                                                                                                                            | 2021–      |
| Advocacy Board Member, MIT Physics Graduate Student Council                                                                                                                                                                                                                                                                                                                                                                                                                            | 2021–2022  |
| Mentor, MIT Graduate & Undergraduate Womxn in Physics                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2021–2024  |
| <b>Outreach</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |
| Invited Talk, MIT List Visual Arts Center<br>Watch here: <a href="https://listart.mit.edu/calendar/graduate-student-talk-megan-masterson">https://listart.mit.edu/calendar/graduate-student-talk-megan-masterson</a>                                                                                                                                                                                                                                                                   | April 2025 |
| Graduate Student Writer, <a href="#">Astrobites</a><br>Find my articles here: <a href="https://astrobites.org/author/mmasterson/">https://astrobites.org/author/mmasterson/</a>                                                                                                                                                                                                                                                                                                        | 2023–2025  |
| Event Organizer, <a href="#">Boston Astronomy on Tap</a>                                                                                                                                                                                                                                                                                                                                                                                                                               | 2023–2024  |
| Member, <a href="#">MIT Sidewalk Astrogazers</a><br>Co-lead of organization during 2023-2024 academic year                                                                                                                                                                                                                                                                                                                                                                             | 2021–      |
| Logistics Officer, Cambridge University Girls in STEM                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2019–2020  |
| Institute of Astronomy Open Evenings Volunteer, University of Cambridge                                                                                                                                                                                                                                                                                                                                                                                                                | 2019–2020  |
| Public Outreach Volunteer, Astrophysics Research Lab at the NC Museum of Natural Sciences                                                                                                                                                                                                                                                                                                                                                                                              | 2017, 2019 |